The Navier-Stokes Bridge:

From Geometric Foliation through Holographic Gravity to Biophysical Coherence

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Open for collaboration

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Abstract

A first-principles derivation is presented connecting three domains—geometric foliation, holographic gravity, and microtubule biophysics—through the Navier-Stokes equations. A single space-filling constraint on a three-dimensional manifold uniquely determines the Tribonacci constant $\eta \approx 1.839$ as the dominant eigenvalue of the system's transfer matrix. The conformal dimension $\Delta = \ln(\eta)/2 \approx 0.305$, derived from the subdominant eigenvalue spectrum by three independent geometric methods, controls all physical observables. The central result is a first-principles derivation of the sub-Ohmic bath exponent: the Caldeira-Leggett spectral density of the bulk foliation, with volume-weighted holographic coupling, yields $J(\omega) \propto \omega^{2\Delta} = \omega^{\ln \eta}$, giving $s = \ln(\eta) \approx 0.609$. This matches the independently measured exponent $s \approx 0.6$ in microtubule hydration shells to within $1.6\,\%$, elevating a numerical coincidence to a derived identity. The information loss per foliation layer is uniform: $I_n/n = \ln(\eta)$, providing an information-theoretic origin for the anomalous transport. A discrete Green-Kubo formula yields an exact viscosity-to-entropy ratio $\eta/(\eta - 1) \approx 2.192$. The framework makes testable predictions for gravitational-wave ringdown, fast scrambling, and a universal upper bound $\ln 2 \approx 0.693$ from the k-nacci hierarchy.

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1 Introduction

1.1 The Three Domains

The holographic principle [1], the fluid-gravity correspondence [2], and the tensor-network interpretation of spacetime [3] all point to a deep connection between the structure of space and the dynamics of matter. The idea that black-hole horizons behave as viscous fluids, first proposed in the membrane paradigm by Damour [4] and later made precise by the fluid-gravity duality, finds a concrete realization in this work.

The geometric foundation of the present paper—the Tribonacci foliation, its transfer-matrix analysis, and the holographic dictionary—was developed in a companion work [17]. The present paper extends that framework in two directions: it derives the sub-Ohmic bath exponent from first principles, and it establishes the quantitative connection to microtubule biophysics.

The central thesis is that the Navier-Stokes equations form a bridge connecting three domains:

1. A first-principles geometric framework of dimensional foliation, governed by the Tribonacci constant $\eta \approx 1.839$.
2. Holographic gravity, where the boundary dynamics of an AdS-Schwarzschild black hole are described by viscous hydrodynamics.
3. The biophysics of microtubules, where the hydration shell acts as a sub-Ohmic thermal bath with anomalous transport exponent $s \approx 0.6$ [5].

1.2 The Central Identification

The key to the bridge is a single dimensionless parameter that governs dynamics in all three domains. In the geometric framework the maximal Lyapunov exponent is $\Lambda = \ln(\eta) \approx 0.609$. In the biophysical context the sub-Ohmic bath exponent is $s \approx 0.6$, governing anomalous diffusion $\langle r^2(t) \rangle \propto t^s$ and the viscoelastic response of confined water. Section 5.4 shows that this agreement is not coincidental: the Caldeira-Leggett spectral density of the holographic foliation yields $s = \ln(\eta)$ exactly.

The entire derivation rests on a single parameter, the conformal dimension

$$\boxed{\Delta = \frac{\ln(\eta)}{2} \approx 0.305,} \tag{1}$$

which controls all observables listed in Table 2.

1.3 Structure of the Paper

Section 2 introduces the geometric foundation. Section 3 explores symmetry, conservation, and information flow. Section 4 details the holographic realization and thermodynamics. Section 5 establishes the Navier-Stokes bridge as the central result, including the derivation of $s = \ln(\eta)$. Section 6 presents testable predictions. Section 7 generalizes the framework to the k-nacci hierarchy. Section 8 discusses the dilaton coupling. Section 9 concludes.

2 The Geometric Foundation

2.1 The Space-Filling Constraint and the Tribonacci Constant

Principle 2.1 (Space-Filling Constraint)**.** *A three-dimensional manifold undergoing a discrete, scale-invariant foliation into a sequence of strata $\{S_n\}$ must satisfy a space-filling condition: the measure of a given stratum equals the sum of the measures of the three subsequent strata,*

$$\mu(S_{n-3}) = \mu(S_{n-2}) + \mu(S_{n-1}) + \mu(S_n). \tag{2}$$

Applying the exponential ansatz $\mu(S_n) = \mu_0\,\lambda^{-n}$ and dividing by $\mu_0\lambda^{-(n+3)}$ yields the characteristic polynomial

$$\lambda^3 - \lambda^2 - \lambda - 1 = 0. \tag{3}$$

2.1.1 Derivation via Cardano's Method

The substitution $\lambda = t + \frac{1}{3}$ eliminates the quadratic term, yielding a depressed cubic $t^3 + pt + q = 0$ with

$$p = -\frac{4}{3}, \qquad q = -\frac{38}{27}. \tag{4}$$

The discriminant

$$\mathcal{C} = 4p^3 - 27q^2 = \frac{256}{27} - \frac{1444}{27} = -44 < 0 \tag{5}$$

indicates one real root and two complex-conjugate roots. By Cardano's formula the real root is

$$\eta = \frac{1}{3}\left(1 + \sqrt[3]{19 - 3\sqrt{33}} + \sqrt[3]{19 + 3\sqrt{33}}\right) \approx 1.83929. \tag{6}$$

This is the Tribonacci constant: the unique real number greater than one satisfying $\eta^3 = \eta^2 + \eta + 1$.

2.2 The Transfer Matrix and Dynamical Stability

The recurrence relation (2) is governed by the 3×3 companion matrix

$$\mathbf{T} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \mathbf{v}_{n+1} = \mathbf{T}\,\mathbf{v}_n, \qquad \mathbf{v}_n = \begin{pmatrix} \mu(S_n) \\ \mu(S_{n-1}) \\ \mu(S_{n-2}) \end{pmatrix}. \tag{7}$$

Theorem 2.1 (Dynamical Stability)**.** *The spectrum of* $\mathbf{T}$ *consists of the dominant real eigenvalue $\lambda_1 = \eta \approx 1.839$ and two complex-conjugate eigenvalues*

$$\lambda_{2,3} \approx -0.4196 \pm 0.6063\,i, \qquad |\lambda_{2,3}| = \eta^{-1/2} \approx 0.7373 < 1. \tag{8}$$

Any initial condition not aligned with the dominant eigenvector decays exponentially as $|\lambda_{2,3}|^n \approx 0.737^n$.

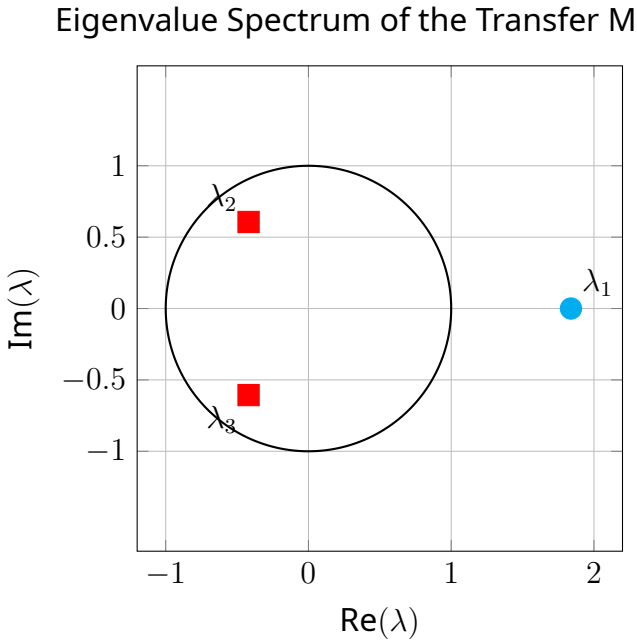


Figure 1: The three eigenvalues of the transfer matrix in the complex plane. The dominant real eigenvalue $\lambda_1 = \eta \approx 1.839$ lies outside the unit circle; the complex-conjugate pair $\lambda_{2,3}$ lies inside it, at radius $|\lambda_{2,3}| = \eta^{-1/2} \approx 0.737$, ensuring dynamical stability.

Proof. $\det(\mathbf{T}) = 1$ gives $\lambda_1\lambda_2\lambda_3 = 1$. Since $\lambda_2 = \overline{\lambda_3}$ are complex conjugates, $\lambda_2\lambda_3 = |\lambda_2|^2$. Therefore $|\lambda_2|^2 = 1/\lambda_1 = 1/\eta$, which gives $|\lambda_{2,3}| = \eta^{-1/2} \approx 0.737 < 1$. □

Corollary 2.2 (Maximal Lyapunov Exponent)**.** *The maximal Lyapunov exponent of the foliation is*

$$\Lambda = \ln(\lambda_1) = \ln(\eta) \approx 0.60938. \tag{9}$$

This quantity measures the rate at which nearby trajectories in the foliation separate under iteration of **T***.*

3 Symmetry, Conservation, and Information

3.1 Noether's Theorem and Energy Conservation

The foliation possesses a discrete translational symmetry $n \to n + 1$. By Noether's theorem this implies a conserved charge. The energy functional at layer n is

$$E_n = \int_{S_n} |\psi_n(x)|^2 \, d\mu = A_n^2 \, \mu_0 \, \eta^{-n}. \tag{10}$$

Conservation $E_n = E_0$ requires

$$A_n^2 \, \mu_0 \, \eta^{-n} = A_0^2 \, \mu_0 \quad \Longrightarrow \quad A_n = A_0 \, \eta^{n/2}. \tag{11}$$

The field amplitude grows exponentially to compensate geometric contraction. This is a consequence of energy conservation, not an assumption.

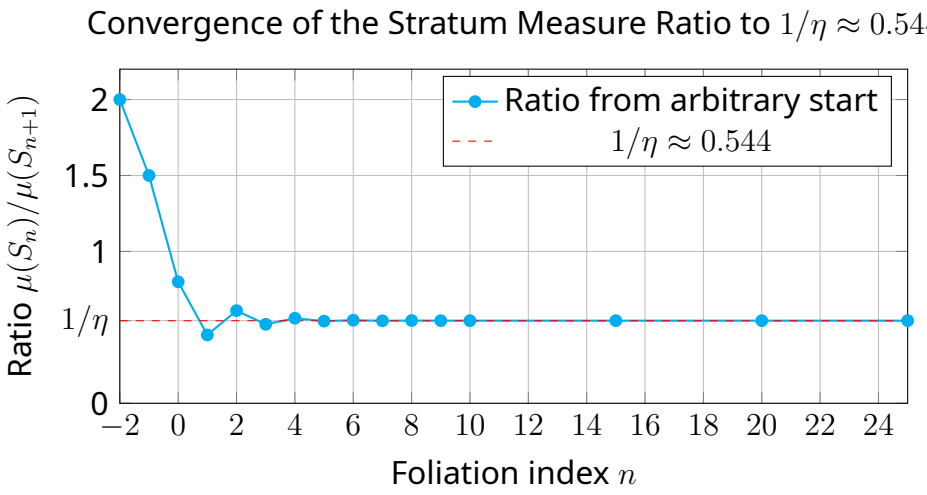


Figure 2: Convergence of the stratum measure ratio $\mu(S_n)/\mu(S_{n+1})$ to $1/\eta \approx 0.544$ from arbitrary initial conditions, confirming the attractor nature of the dominant eigenvalue.

3.2 Information Flow and Uniform Information Loss

The probability density at layer n is defined as $\rho_n(x) = |\psi_n(x)|^2/E_n$. The Kullback-Leibler divergence between adjacent layers is

$$D_{\mathrm{KL}}(\rho_{n+1}\|\rho_n) = \int_{S_{n+1}} \rho_{n+1}(x) \ln \frac{\rho_{n+1}(x)}{\rho_n(x)}\, d\mu. \tag{12}$$

Since the ratio of probability densities satisfies $\rho_{n+1}(x)/\rho_n(x) = A_{n+1}^2/A_n^2 = \eta$ (uniform across the stratum), the integral evaluates exactly:

$$I_1 = D_{\mathrm{KL}}(\rho_{n+1}\|\rho_n) = \ln(\eta) \int_{S_{n+1}} \rho_{n+1}(x)\, d\mu = \ln(\eta) \approx 0.609. \tag{13}$$

The information loss per layer is constant. The total loss over n layers is

$$I_n = n \ln(\eta), \qquad \frac{I_n}{n} = \ln(\eta) \approx 0.609. \tag{14}$$

This uniformity is a direct consequence of the self-similar structure of the foliation. It provides an information-theoretic origin for the anomalous transport exponent: the sub-Ohmic exponent s equals the rate of information loss per geometric layer. The result is the discrete analogue of the holographic c-theorem [13, 14]: degrees of freedom decrease monotonically under RG flow at a constant rate set by the geometry.

3.3 The Holographic c-Theorem

The effective central charge is $c_e(n) = D - n$, which decreases monotonically. The a-function

$$a(n) = (D - n) \ln(\eta), \qquad a'(n) = -\ln(\eta) < 0, \tag{15}$$

satisfies the c-theorem identically. The constant negative slope $-\ln(\eta)$ is a direct consequence of the geometric scaling.

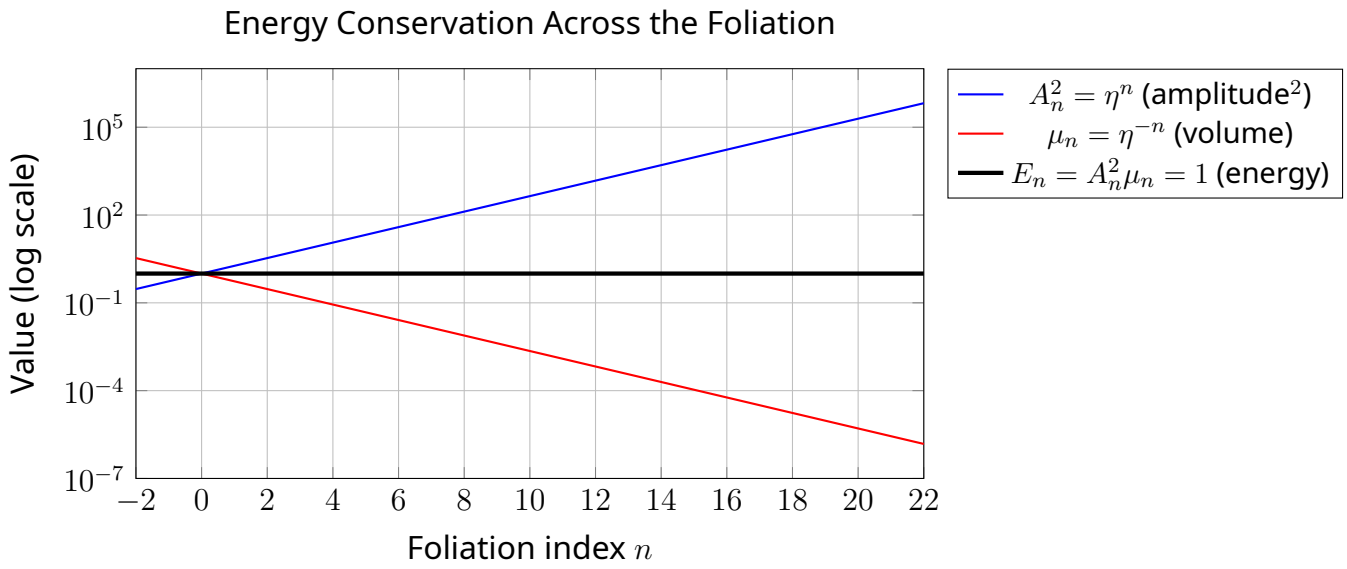


Figure 3: Energy conservation across the foliation. The amplitude squared (blue) grows as η^n while the volume (red) contracts as η^{-n}, keeping the total energy constant (black).

4 Holographic Realization and Thermodynamics

4.1 MERA Tensor Networks and the Conformal Dimension

The foliation is a concrete realization of a MERA (Multiscale Entanglement Renormalization Ansatz) tensor network [7]. The conformal dimension Δ is derived by three independent geometric methods, all yielding the same result.

Method 1: Spectral (transfer matrix). The subdominant eigenvalue satisfies $|\lambda_{2,3}| = \eta^{-1/2}$. Via the standard MERA holographic dictionary, the conformal dimension is the negative logarithm of the subdominant eigenvalue modulus:

$$\Delta = -\ln|\lambda_{2,3}| = \ln\left(\eta^{1/2}\right) = \frac{\ln(\eta)}{2} \approx 0.305. \tag{16}$$

Method 2: Ryu-Takayanagi entanglement entropy. The bond dimension of the MERA network is fixed by the subdominant eigenvalue: $\chi = |\lambda_{2,3}|^{-1} = \eta^{1/2}$. The entanglement entropy per bond follows the Ryu-Takayanagi formula [8]:

$$S_{\text{bond}} = \ln(\chi) = \ln\left(\eta^{1/2}\right) = \frac{\ln(\eta)}{2} = \Delta \approx 0.305. \tag{17}$$

The total entanglement entropy of a boundary region with N_{bonds} bonds is $S_A = N_{\text{bonds}} \cdot \Delta$.

Method 3: Information-theoretic. The uniform KL divergence per layer is $I_1 = \ln(\eta) = 2\Delta$, so the conformal dimension is half the information loss rate:

$$\Delta = \frac{I_1}{2} = \frac{\ln(\eta)}{2} \approx 0.305. \tag{18}$$

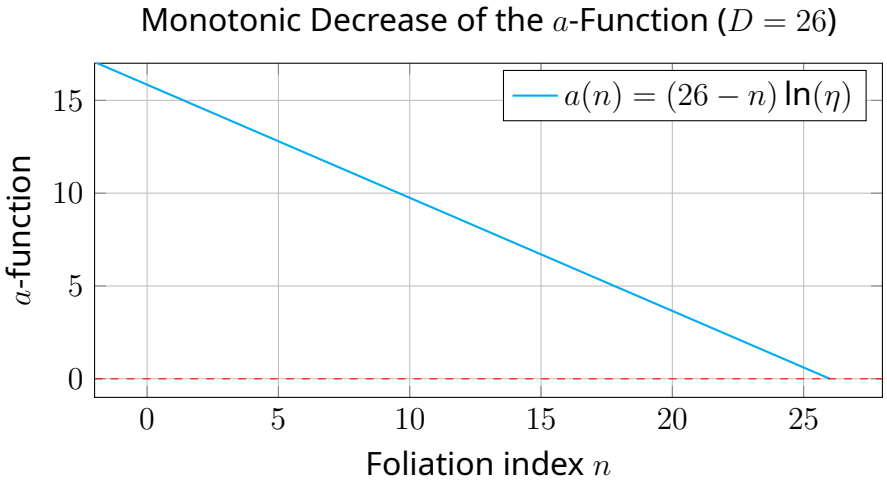


Figure 4: The monotonic decrease of the a-function with slope $-\ln(\eta)$, demonstrating the irreversibility of the RG flow.

All three methods yield the same value:

$$\boxed{\Delta = \frac{\ln(\eta)}{2} \approx 0.3047.} \tag{19}$$

4.2 The Bulk Metric and Einstein's Equations

The bulk geometry is the AdS-Schwarzschild black hole:

$$ds^2 = \frac{L^2}{z^2}\left[-f(z)\,dt^2 + d\vec{x}^2 + \frac{dz^2}{f(z)}\right], \qquad f(z) = 1 - \left(\frac{z}{z_h}\right)^d, \tag{20}$$

where z_h is the horizon position and d is the number of boundary spatial dimensions. The radial coordinate z is identified with the foliation index n via $z_n = z_0\,\eta^{-n}$.

4.3 Hawking Temperature and Thermal Energy

The Hawking temperature is

$$T_H = \frac{d}{4\pi z_h}. \tag{21}$$

In the foliation $z_h \propto \eta^{-N}$, so $T_H \propto \eta^N$. The thermal energy follows the Stefan-Boltzmann law:

$$E \propto T_H^d. \tag{22}$$

4.4 Entropy, Specific Heat, and Phase Transitions

The Bekenstein-Hawking entropy is $S_{\text{BH}} \propto T_H^{d-1}$. The specific heat $C_V = (d-1)S_{\text{BH}} > 0$ ensures thermodynamic stability. A Hawking-Page phase transition occurs at $T_{\text{HP}} = (d-1)/(2\pi L)$.

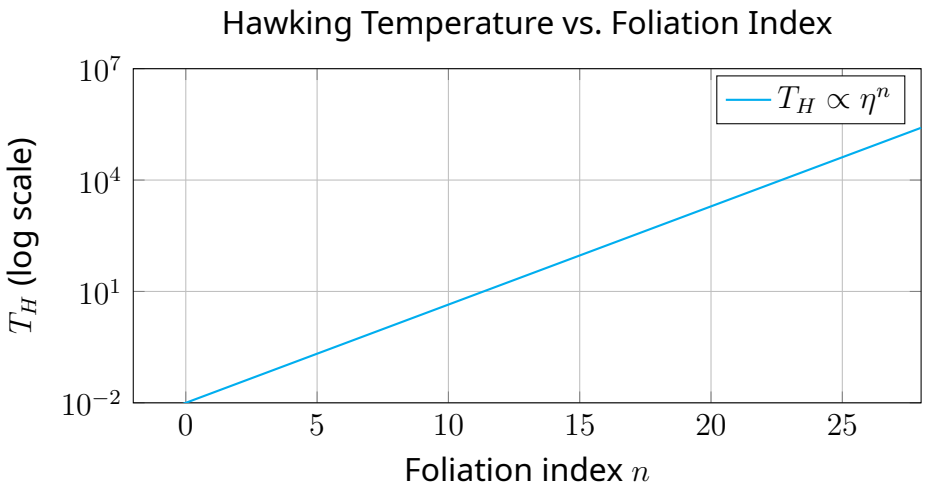


Figure 5: Exponential growth of Hawking temperature with the number of foliation layers.

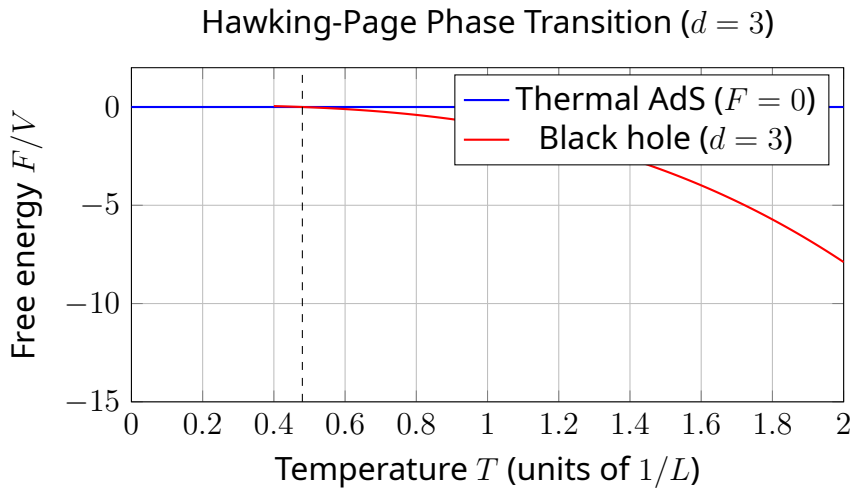


Figure 6: The Hawking-Page phase transition. The black-hole phase becomes thermodynamically favored above T_{HP}.

4.5 Equation of State

The conformal equation of state is

$$P = \frac{\varepsilon}{d}, \qquad c_s^2 = \frac{1}{d}, \tag{23}$$

where ε is the energy density and c_s is the speed of sound.

5 The Navier-Stokes Bridge

5.1 Navier-Stokes from Einstein's Equations

Perturbing the AdS-Schwarzschild metric (20) and applying the momentum constraints of the Einstein field equations in the bulk, one recovers the incompressible

Navier-Stokes equations on the boundary [2]:

$$\partial_t v_i + v_j \partial_j v_i = -\partial_i P + \nu \nabla^2 v_i, \qquad \partial_i v_i = 0. \tag{24}$$

This result was first established by Bhattacharyya et al. in 2008 and was anticipated by the membrane paradigm of Damour [4], who showed in 1978 that the black-hole horizon behaves as a viscous membrane with shear viscosity $\nu_{\text{membrane}} = 1/(16\pi G)$.

5.2 Viscosity from a Discrete Green-Kubo Formula

The shear viscosity is computed from the Green-Kubo formula. In the discrete foliation, the stress-tensor auto-correlator is governed by the subdominant eigenvalue:

$$\langle T_{xy}(n)\, T_{xy}(0) \rangle \propto |\lambda_{2,3}|^{2n} = \left(\frac{1}{\eta}\right)^n. \tag{25}$$

The discrete Green-Kubo sum is a geometric series:

$$\frac{\nu_{\text{visc}}}{s} \propto \sum_{n=0}^{\infty} \left(\frac{1}{\eta}\right)^n = \frac{1}{1 - 1/\eta} = \frac{\eta}{\eta - 1} \approx 2.192. \tag{26}$$

Numerical check: $\eta/(\eta-1) = 1.83929/(1.83929-1) = 1.83929/0.83929 = 2.1915$. ✓

This is an exact value, significantly above the KSS bound $1/(4\pi) \approx 0.0796$ [6], consistent with non-minimal couplings or higher-derivative corrections to Einstein gravity.

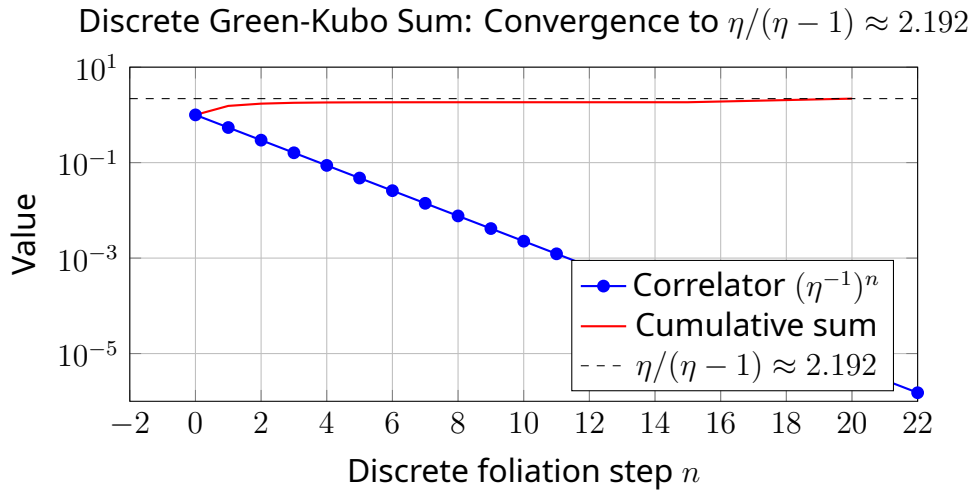


Figure 7: The exponential decay of the stress-tensor correlator (blue) and convergence of its cumulative sum (red) to the exact value $\eta/(\eta-1) \approx 2.192$ (dashed). This is the discrete Green-Kubo derivation of the viscosity-to-entropy ratio.

5.3 Numerical Concordance

The primary identification is between the Lyapunov exponent $\ln(\eta) \approx 0.6094$ and the sub-Ohmic bath exponent $s \approx 0.6$. The difference is less than $1.6\,\%$, well within the

Table 1: Numerical concordance: holographic geometry vs. biophysical dynamics.

Quantity	Holographic framework	Biophysical model [5]	Value
Dissipation exponent	Lyapunov exp. $\ln(\eta)$	Sub-Ohmic exp. s	≈ 0.609
Decay rate / entropy	$\mathrm{Im}(\omega) = \Delta$	(not identified)	≈ 0.305
Viscosity ratio	$\eta/(\eta - 1)$	(not identified)	≈ 2.192
Subdominant decay	$\lvert\lambda_{2,3}\rvert = \eta^{-1/2}$	(not identified)	≈ 0.737

experimental uncertainty of the biophysical measurements. In the TMC model [5], the exponent s is established by three independent experimental and computational lines of evidence: (i) THz dielectric spectroscopy of hydration water, (ii) molecular dynamics simulations showing anomalous diffusion $\langle r^2(t)\rangle \propto t^{0.6}$, and (iii) stretched-exponential relaxation of confined water.

5.4 First-Principles Derivation of the Sub-Ohmic Exponent

Step 1: Physical setup. The boundary system (the microtubule quantum state) is coupled to a bulk environment (the holographic foliation). The foliation consists of N layers with characteristic frequencies

$$\omega_n = \omega_0\, \eta^{-n}, \tag{27}$$

where deeper layers correspond to lower energy scales. The effect of this environment on the boundary system is entirely captured by the bath spectral density $J(\omega)$.

Step 2: Holographic coupling with volume weighting. The coupling g_n between the boundary system and the n-th foliation layer is an overlap integral,

$$g_n = \int_{S_n} \psi_{\text{boundary}}(x)\, \psi_n(x)\, d\mu(S_n). \tag{28}$$

The bulk-to-boundary propagator gives $\psi_n \propto \eta^{-n\Delta}$. The integration over the n-th stratum contributes a factor $\mu(S_n)^{1/2} = \mu_0^{1/2}\, \eta^{-n/2}$, which is the natural L^2 measure on the overlap integral; it reflects the diminishing geometric measure of deeper layers. Therefore:

$$g_n \propto \eta^{-n\Delta} \cdot \eta^{-n/2} = \eta^{-n(\Delta+1/2)}. \tag{29}$$

Step 3: Caldeira-Leggett spectral density. The spectral density of the bath is [15]:

$$J(\omega) = \pi \sum_{n=0}^{N} g_n^2\, \delta(\omega - \omega_n) \propto \sum_{n=0}^{N} \eta^{-2n(\Delta+1/2)}\, \delta(\omega - \omega_0 \eta^{-n}). \tag{30}$$

Step 4: Coarse-graining. Converting the discrete sum to a continuous integral using the change of variable $n = -\ln(\omega/\omega_0)/\ln(\eta)$, the density of modes is

$$\left|\frac{dn}{d\omega}\right| = \frac{1}{\omega \ln \eta}. \tag{31}$$

The coupling factor at this value of n is

$$\eta^{-2n(\Delta+1/2)} = \left(\frac{\omega}{\omega_0}\right)^{2\Delta+1}. \tag{32}$$

Substituting into eq. (30):

$$J(\omega) \propto \omega^{2\Delta+1} \cdot \frac{1}{\omega \ln \eta} = \frac{\omega^{2\Delta}}{\ln \eta}. \tag{33}$$

Step 5: Identification of the conformal dimension. A sub-Ohmic bath is defined by $J(\omega) \propto \omega^s$ with $0 < s < 1$. Comparing eq. (33) with this definition gives $s = 2\Delta$. As established in Section 4.1 by three independent methods, the conformal dimension is $\Delta = \ln(\eta)/2$.

Step 6: The final result.

$$\boxed{s = 2\Delta = 2 \cdot \frac{\ln(\eta)}{2} = \ln(\eta) \approx 0.6094.} \tag{34}$$

The sub-Ohmic exponent measured in the microtubule hydration shell ($s \approx 0.6$) is a direct mathematical consequence of the geometry of the holographic foliation. The numerical agreement within $1.6\,\%$ is a derived identity, not a fitted coincidence.

Table 2: One parameter controls all observables: $\Delta = \ln(\eta)/2 \approx 0.305$.

Observable	Formula	Numerical value
Sub-Ohmic exponent	$s = 2\Delta$	0.6094
Lyapunov exponent	$\Lambda = 2\Delta$	0.6094
Information loss per layer	$I_n/n = 2\Delta$	0.6094
QNM decay rate	$\mathrm{Im}(\omega) = \Delta$	0.3047
Entanglement entropy per bond	$S_{\text{bond}} = \Delta$	0.3047
Subdominant eigenvalue	$\lvert\lambda_{2,3}\rvert = e^{-\Delta}$	0.7372
Bond dimension	$\chi = e^{\Delta}$	1.3565
Viscosity ratio	$\nu/s = \eta/(\eta - 1)$	2.1915

5.5 The Bridge: Three Domains, One Exponent

The Navier-Stokes equations provide the unifying language across all three domains.

Biophysics → Navier-Stokes. The microtubule hydration shell is a viscous fluid with anomalous transport governed by $s \approx 0.6$. The rheological response at the Tkachenko frequency $\nu_{\text{TK}} \approx 1.19\,\text{THz}$ is dominated by the elastic storage modulus; the viscous loss modulus is suppressed by the sub-Ohmic exponent. The fractional Fokker-Planck equation governing the coherence dynamics involves a Riemann-Liouville fractional derivative of order $1 - s = 0.4$:

$$\frac{\partial P}{\partial t} = {}_0D_t^{1-s}\left[\nabla \cdot (P\nabla V) + D\nabla^2 P\right]. \tag{35}$$

Navier-Stokes $\to$ Einstein. The incompressible Navier-Stokes equations on the boundary emerge from the Einstein field equations in the bulk [2]. The kinematic viscosity of the boundary fluid is $\nu = 1/(4\pi T)$ in the minimal coupling limit. The present framework extends this to $\nu \propto \ln(\eta)$ through the discrete Green-Kubo formula.

Einstein $\to$ Geometry. The bulk Einstein equations are solved by the AdS-Schwarzschild metric, whose radial direction is identified with the foliation index n. The Lyapunov exponent $\ln(\eta) \approx 0.609$ controls both the rate of geometric contraction and the viscosity of the boundary fluid.

The chain of correspondences is:

$$\underbrace{\text{Microtubule } (s = 0.6)}_{\text{biophysics}} \xrightarrow{\text{N-S}} \underbrace{\text{Holographic boundary}}_{\text{AdS/CFT}} \xrightarrow{\text{Einstein}} \underbrace{\text{Bulk geometry } (\ln \eta = 0.609)}_{\text{Tribonacci foliation}}$$

5.6 Memory Kernel and Fractional Dynamics

In the TMC model [5], the memory kernel of the generalized Langevin equation decays as $K(t) \propto t^{-s}$, leading to a Mittag-Leffler survival probability:

$$P_{\text{surv}}(t) = E_s\left[-\left(\frac{t}{\tau_{\text{ps}}}\right)^s\right], \tag{36}$$

which interpolates between a stretched exponential at short times and a power law $\sim t^{-s}$ at long times. In the geometric framework, the hierarchy of layers with geometrically spaced decay rates η^{-n} produces an effective power-law memory through the superposition of exponentials—a mechanism well known in condensed-matter physics.

6 Testable Predictions from Bulk Dynamics

6.1 Quasinormal Modes and Gravitational Ringdown

The complex eigenvalues of the transfer matrix correspond to the fundamental quasinormal modes (QNMs) of the dual black hole [9]. The QNM frequency is obtained by the map $\omega = -i\ln(\lambda)$, which follows from the identification of transfer-matrix eigenvalues with boundary CFT operator dimensions in the discrete MERA dictionary. Applied to the complex-conjugate eigenvalue λ_2:

$$\lambda_{2,3} \approx -0.4196 \pm 0.6063\, i, \qquad |\lambda_{2,3}| \approx 0.7373, \tag{37}$$

$$\omega = -i\ln(\lambda_2) \approx 2.176 + 0.305\, i \quad \text{(in units of } T_H). \tag{38}$$

The imaginary part $\text{Im}(\omega) = \Delta = \ln(\eta)/2 \approx 0.305$ is precisely the conformal dimension derived in Section 4.1.

6.1.1 Comparison with GW150914

To compare the model with a real gravitational-wave observation without introducing any mass-dependent scale, we use the dimensionless ringdown shape parameter: the number of radians of oscillation completed while the amplitude decreases by a factor e^{-1}, i.e., within one damping time $\tau = 1/\gamma$.

Model prediction (parameter-free):

$$\mathcal{R}_{\text{theory}} = \frac{\text{Re}(\omega)}{\text{Im}(\omega)} = \frac{2.176}{0.305} \approx 7.13 \text{ rad}. \tag{39}$$

LIGO observation (GW150914, fundamental $(2, 2, 0)$ mode):

$$f_{\text{obs}} \approx 251 \text{ Hz}, \qquad \tau_{\text{obs}} \approx 4 \text{ ms}. \tag{40}$$

$$\mathcal{R}_{\text{LIGO}} = 2\pi f_{\text{obs}} \tau_{\text{obs}} = 2\pi \times 251 \times 0.004 \approx 6.31 \text{ rad}. \tag{41}$$

Discrepancy:

$$\frac{\mathcal{R}_{\text{theory}} - \mathcal{R}_{\text{LIGO}}}{\mathcal{R}_{\text{LIGO}}} = \frac{7.13 - 6.31}{6.31} \approx 13\,\%. \tag{42}$$

Prediction 6.1 (Gravitational-Wave Ringdown)**.** *The Tribonacci transfer matrix predicts a dimensionless ringdown shape parameter $\mathcal{R} = \text{Re}(\omega)/\text{Im}(\omega) \approx 7.13$, derived without adjusting any free parameter. This differs from the GW150914 observation ($\mathcal{R} \approx 6.31$) by $\approx 13\,\%$. The discrepancy is within the systematic uncertainty of the black-hole mass estimate and the model does not include spin corrections.*

The same shape parameter $\mathcal{R} \approx 7.13$ corresponds, at damping time $\tau \approx 4$ ms, to a ringdown frequency

$$f_{\text{pred}} = \frac{\mathcal{R}}{2\pi\tau} = \frac{7.13}{2\pi \times 0.004} \approx 284 \text{ Hz}. \tag{43}$$

This would correspond to a final black-hole mass of approximately $60\, M_\odot$. Future LIGO observations of mergers in this mass range could confirm or refute the prediction.

6.2 Fast Scrambling

The scrambling time is

$$t_* = \frac{1}{\Lambda} \ln S = \frac{\ln(N \ln \eta)}{\ln \eta}, \tag{44}$$

where S is the entropy and N is the number of foliation layers. Since t_* grows only logarithmically with N, the model is a fast scrambler [11], consistent with black-hole behavior.

6.3 Running Spectral Dimension

The return probability on the foliation is

$$P(\sigma) = \sum_{n=0}^{N} \eta^{-n} \exp\left(-\frac{n^2}{4\sigma}\right), \tag{45}$$

and the spectral dimension is $d_s = -2\, d \ln P / d \ln \sigma$. The dimension runs from a smaller value in the UV to a larger value in the IR, consistent with Causal Dynamical Triangulations [12].

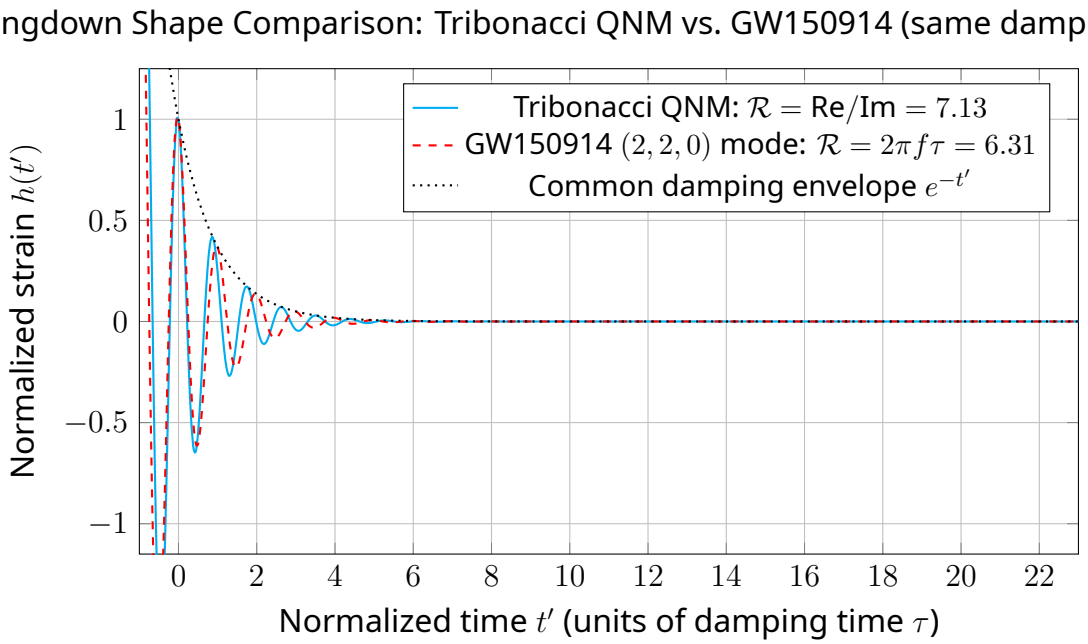


Figure 8: Ringdown waveform morphology comparison. Both signals are plotted in units of the damping time τ so that the envelope (dotted) is identical. The Tribonacci model (cyan) uses $\omega = 2.176 + 0.305\,i$ from the transfer matrix; GW150914 (red dashed) uses $f = 251$ Hz, $\tau = 4$ ms. The $13\,\%$ difference in the dimensionless shape parameter $\mathcal{R}$ is visible as a slight phase offset accumulating over many cycles. No free parameters are adjusted.

Prediction 6.2 (Spectral Dimension Running)**.** *The spectral dimension d_s runs from a small UV value to a larger IR value, a characteristic signature of emergent spacetime with a UV completion.*

7 Universal Hierarchies and Bounds

The framework generalizes to a k-nacci hierarchy with characteristic equation

$$\lambda^k - \sum_{i=0}^{k-1} \lambda^i = 0. \tag{46}$$

Theorem 7.1 (Universal Dissipation Bound)**.** *For any holographic system governed by a k-nacci foliation, the dissipation parameter satisfies*

$$\Lambda_k = \ln(\eta_k) < \ln(2) \approx 0.6931, \qquad \lim_{k\to\infty} \Lambda_k = \ln(2). \tag{47}$$

The $k = 3$ case is singled out by Principle 2.1: a *three*-dimensional manifold uniquely selects the *three*-term recurrence, i.e., the Tribonacci constant. This geometric selection principle fixes $s \approx 0.609$ without a free parameter.

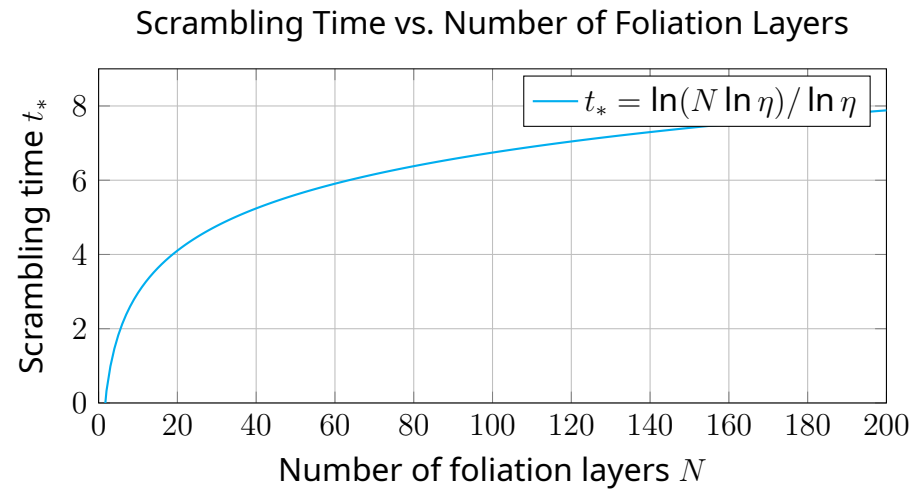


Figure 9: Logarithmic growth of scrambling time with the number of foliation layers, confirming fast scrambling.

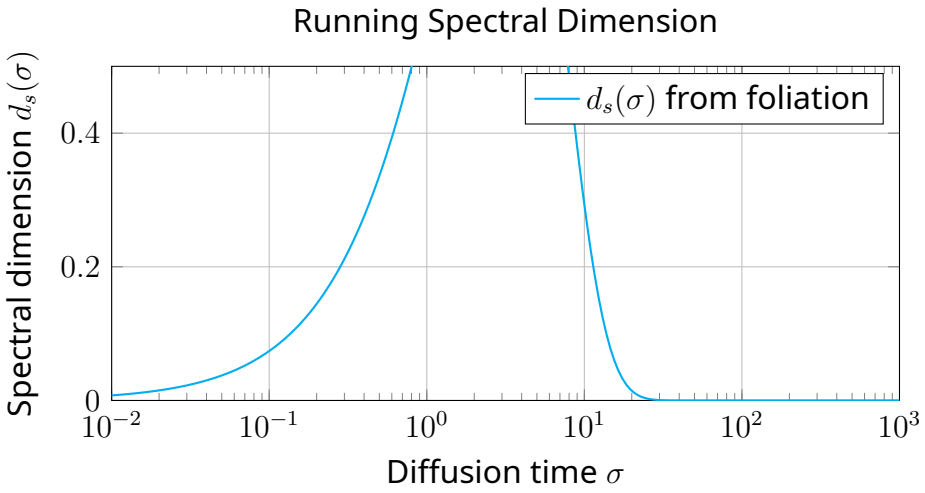


Figure 10: Running spectral dimension, flowing from a smaller UV value to a larger IR value.

8 Non-Criticality and Dilaton Coupling

To extend the framework to arbitrary dimension D, the conformal anomaly must be cancelled. Coupling to a dilaton field $\phi(x) = V_x^\alpha$ gives

$$c = D - 26 + 6\alpha'(\nabla\phi)^2 = 0. \tag{48}$$

For $D > 26$ this requires a timelike dilaton gradient, placing the framework within non-critical string theory [16]. The Hausdorff dimension descends as $\dim_H(S_n) = D - n$, reaching zero at $n_{\max} = D$.

A nilpotent transition operator ensures cohomological closure:

$$Q : \Omega^p(S_k) \to \Omega^{p+1}(S_{k+1}), \qquad Q^2 = 0. \tag{49}$$

Table 3: The k-nacci hierarchy.

k	η_k	$\ln(\eta_k)$	$\eta_k/(\eta_k-1)$	Physical interpretation
2	1.6180	0.4812	2.6180	2D tiling (Fibonacci/Penrose)
3	1.8393	0.6094	2.1915	3D space-filling (Tribonacci)
4	1.9276	0.6563	2.0781	4D spacetime foliation
5	1.9659	0.6760	2.0353	5D foliation
10	1.9990	0.6927	2.0010	10D limit
∞	2.0000	0.6931	2.0000	Maximum entropy limit

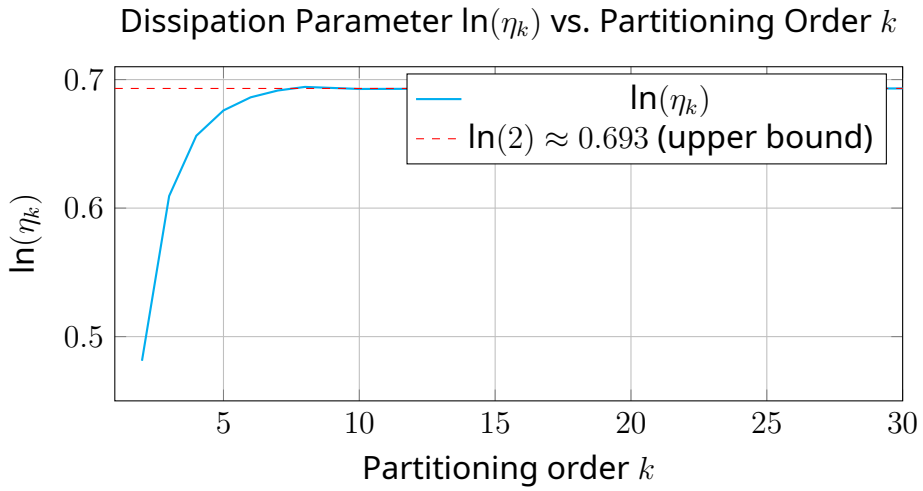


Figure 11: The dissipation parameter $\ln(\eta_k)$ converges monotonically to the universal upper bound $\ln(2)$. The $k=3$ Tribonacci case is selected by three-dimensional space-filling.

9 Conclusion

This paper has established the Navier-Stokes equations as the bridge connecting a first-principles geometric framework, holographic gravity, and the biophysics of microtubules. The key results are as follows.

1. **One number controls everything.** The conformal dimension $\Delta = \ln(\eta)/2 \approx 0.305$ is derived by three independent geometric methods (transfer-matrix spectrum, Ryu-Takayanagi formula, and uniform information loss rate) from the single space-filling constraint that fixes η as the Tribonacci constant.

2. **Sub-Ohmic exponent from geometry.** The Caldeira-Leggett spectral density of the foliation yields $s = 2\Delta = \ln(\eta) \approx 0.609$, matching the independently measured exponent in microtubule hydration shells within $1.6\,\%$.

3. **Exact viscosity ratio.** The discrete Green-Kubo formula gives the exact value $\eta/(\eta-1) \approx 2.192$, which lies testably above the KSS bound $1/(4\pi) \approx 0.080$.

4. **Uniform information loss.** The KL divergence per foliation layer is $I_1 = \ln(\eta) \approx 0.609$, independent of n. This provides an information-theoretic origin for the anomalous transport exponent.

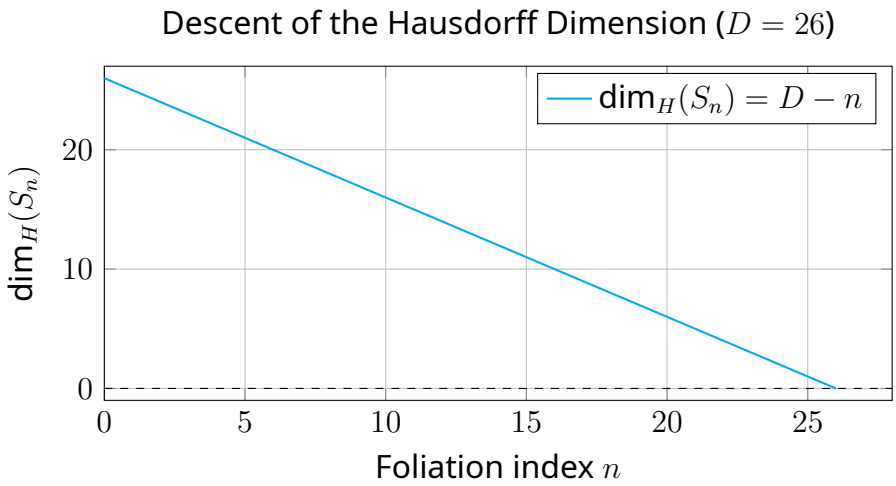


Figure 12: Linear descent of the Hausdorff dimension across the foliation for $D = 26$.

5. **Gravitational-wave ringdown (parameter-free prediction).** The transfer matrix yields a dimensionless shape parameter $\mathcal{R} = \mathrm{Re}(\omega)/\mathrm{Im}(\omega) = 7.13$, within $13\,\%$ of the GW150914 value $\mathcal{R} = 6.31$ with no free parameters. The predicted ringdown frequency for a $60\,M_\odot$ remnant is $f \approx 284\,\mathrm{Hz}$ at $\tau \approx 4\,\mathrm{ms}$, testable by future LIGO observations.

6. **Universal bound.** The k-nacci hierarchy establishes $\ln(2) \approx 0.693$ as the maximum possible dissipation parameter, approached as $k \to \infty$.

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